Cell processes

Copy the instructions

1 Chromosomes & genes

Nucleus contains **DNA chromosomes,** each with many genes; normally paired in body cells.

After copying, each X is **one chromosome** with two joined identical copies (sister chromatids).

2 Grow & copy DNA



Before copying Joined copies

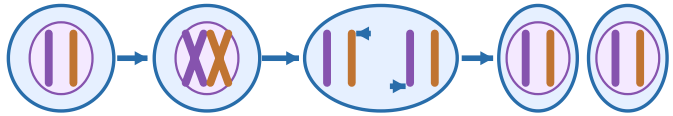
Example shown: **two chromosomes.**

Before mitosis: cell grows; more ribosomes and mitochondria form.

DNA replicates. **DNA amount doubles; chromosome number does not.** Joined copies will separate.

Divide & differentiate

3 Mitosis



Before Copy Separate 2 cells

Example shown: **two chromosomes.**

Joined copies **separate;** one **complete set** moves to each end. The nucleus divides.

Copying happened **before** mitosis. Chromosome number is maintained, not halved.

4 Divide the cell

Cytoplasm and membrane divide → **two genetically identical daughter cells,** with the parent's chromosome number.

For growth, development, repair/replacement, e.g. healing skin. Bacteria use **binary fission,** not mitosis.

5 Human stem cells

Undifferentiated cells divide and produce cells that differentiate.

Embryonic: most human types. **Adult bone marrow:** limited range, including blood cells. Source determines the range.

Use stem cells

6 Medical uses

Potential replacement of insulin-producing cells in **diabetes,** or nerve cells in **paralysis.**

Therapeutic cloning: embryo has patient's genes → matched tissue is not rejected by their immune system. Potential, not guaranteed cures.

7 Benefits, risks & ethics

Benefit: repair damaged tissue. Risks: **viral infection** and uncontrolled division.

Embryo destruction raises ethical/religious objections; others prioritise treatment benefits. Weigh source, benefits and risks; give a **reasoned judgement.**

8 Plant meristems

Can form **any plant cell type throughout life.** Produce clones quickly and economically.

Clone rare plants to prevent extinction; useful crops for features such as disease resistance. Plant cells come from plant meristems, not human stem cells.